

$$\lim_{x \rightarrow 0} \exp\left(\frac{1}{x^2 \ln x}\right) \quad \log \left(\frac{x^4 \sqrt[3]{\sin x - x^2 x + x^4}}{e^{x^2} - \cos x} \right)$$

$\sim x^3$ $\sim x^2$ $\frac{2}{9} x^2 = \frac{x^4/3}{3/2 x^2}$

$$x - \sin^2 x = x^2 - \left(x - \frac{x^3}{6} + o(x^3) \right)$$

$$= x^2 - x^2 + \frac{x^4}{3} + \frac{x^6}{36} + o(x^6) + o(x^4)$$

$$= \frac{x^4}{3} + o(x^4)$$

$$\cancel{x^4 \sqrt[3]{\sin x}} + \frac{x^4}{3} + o(x^4) = \frac{x^4}{3} + o(x^4)$$

$$\frac{x^4 \sqrt[3]{\sin x}}{x^4} = \sqrt[3]{\sin x} \rightarrow 0 \Rightarrow \frac{x^4 \sqrt[3]{\sin x}}{x^4} = o(x^4)$$

$$e^{x^2} - \cos x$$

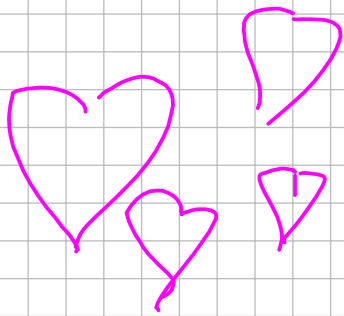
$$e^t = 1 + t + \frac{t^2 + o(t^2)}{2} x^2 \Rightarrow e = 1 + x^2 + \frac{x^4}{2} + o(x^4)$$

x^2

$$O - \cos x = 1 + x^2 + \frac{x^4}{2} - \left(1 - \frac{x^2}{2} + \frac{x^4}{4}\right) + o(x^4)$$

$$= \frac{3}{2}x^2 + o(x^2)$$

primo termine
significazione
NON SERVE
ALTRO



$$\int_{-3}^b \frac{(t^2 + 4t - 1)e^t}{u} dt$$

u (green) v' (orange)